

Game Proposal: “Dear Mother”

CPSC 427 – Video Game Programming

Team: 427 Rogues

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Story:

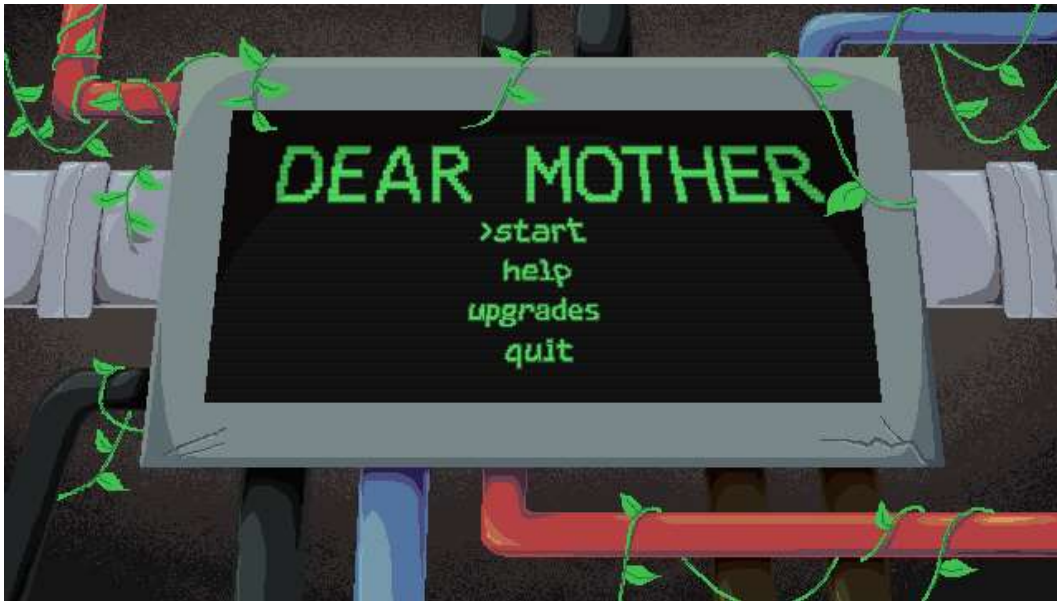
After advancements in technology created AI that strayed from its original purpose of assisting humanity, the world's resources and life were slowly consumed by four autonomous robot factories called Mothers. The ground-level mother is the Mother of Preservation. She was corrupted to freeze herself and not intervene with the world, as preserving everything is an impossible task. Beneath the Mother of Preservation was the Mother of Dispersion, tasked with Dispersing Robots generated by the lower factories across the world. She started stealing other factories' robots and making them her own, using them as a personal army. Below her was the Mother of Recycling, who was corrupted to recycle life that was still alive. The bottom Mother was the mother of Propagation, tasked with creating robots. She was corrupted to run the world out of resources by creating veritable armies of robots. Each mother lays claim to one of the four floors of a massive industrial complex that burrows deep into the earth. Depravity and decay reign until the Mother of Preservation's core directive is re-initialized, turning her against her kin.

This Mother creates you—our player—to venture deep into the complex and destroy the other three mothers. You play as iterations of a robot, finding and equipping useful items from enemies you've vanquished on the floors below. You have a limited number of item slots, so you will have the option to scrap these items. This scrap can be sent up to your Mother after successfully killing a Mother or traded with NPC traders in-run for items or to regain health. This scrap, if sent up, can be used in between runs to permanently upgrade your Mother's factory which improves the features of the next iteration of yourself. You will traverse rooms of equal size until you find the mother's room, which is a larger room where a boss fight takes place. After defeating the mother, an elevator door will open to allow you to descend to the next floor.

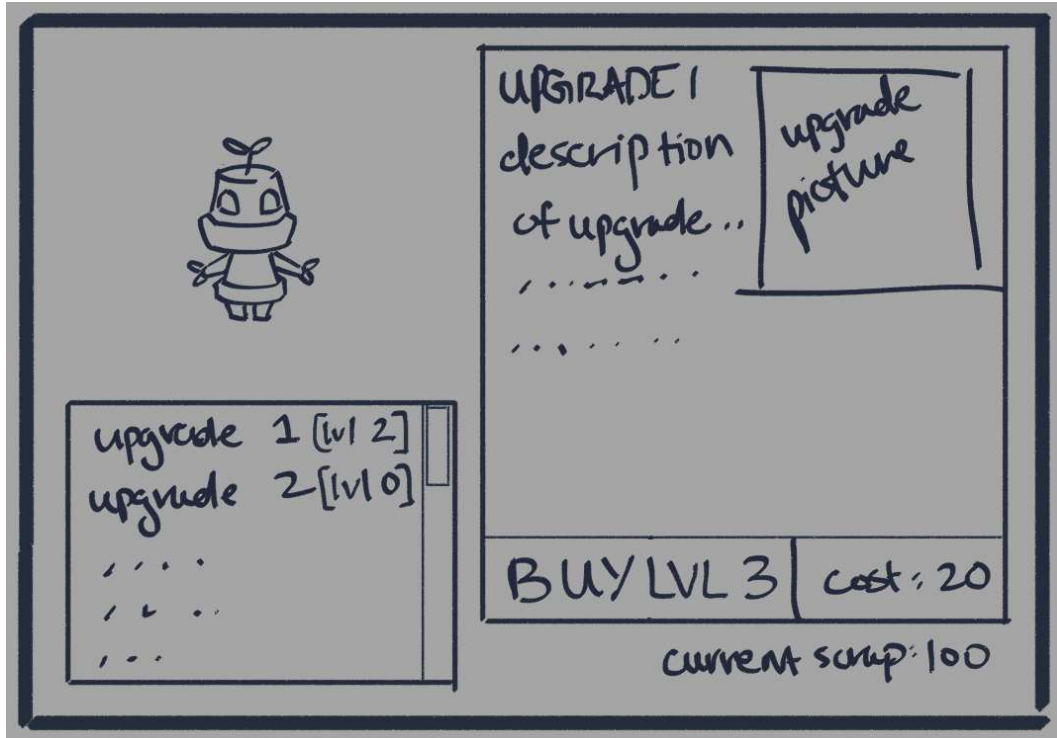
As you delve deeper into the bowels of the factory, foes become increasingly out of touch with the natural world, aggressive and corrupted by the twisted core directives of the mothers that created them. This creates a ramping difficulty. The three hostile mothers will be

different (assets, attacks, etc). You will find that the complex is so large that every time you venture out, you find a different path to descend toward its core. Occasionally, you may stumble upon dead iterations of yourself from which you can recover scrap, robots who have abandoned their mothers (NPC traders) and other interesting rooms.

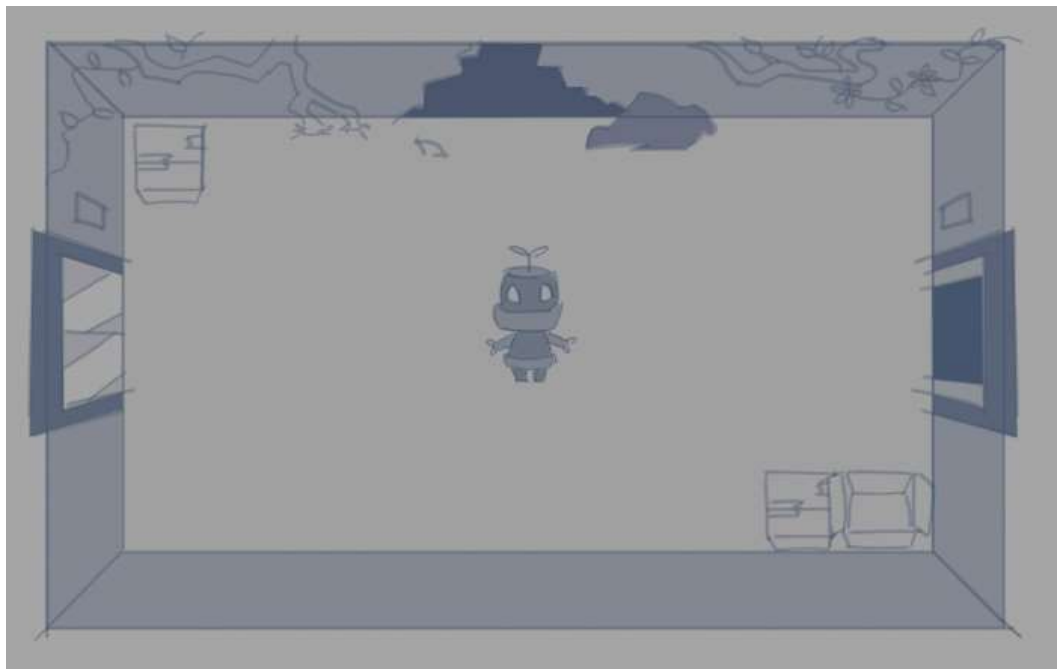
Scenes:



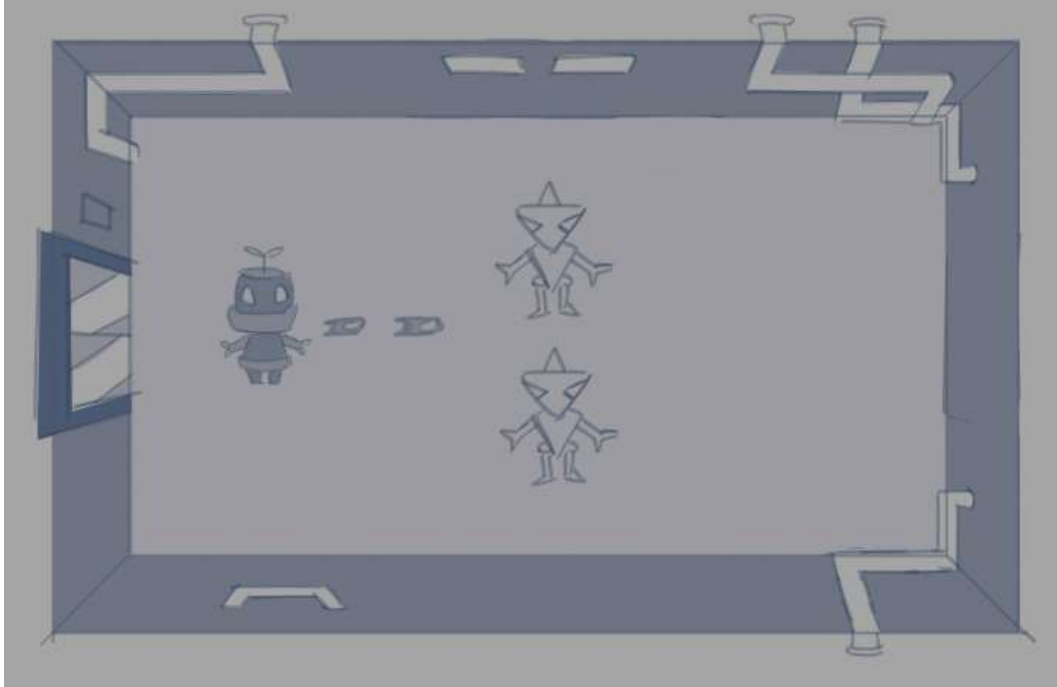
Start Screen Concept



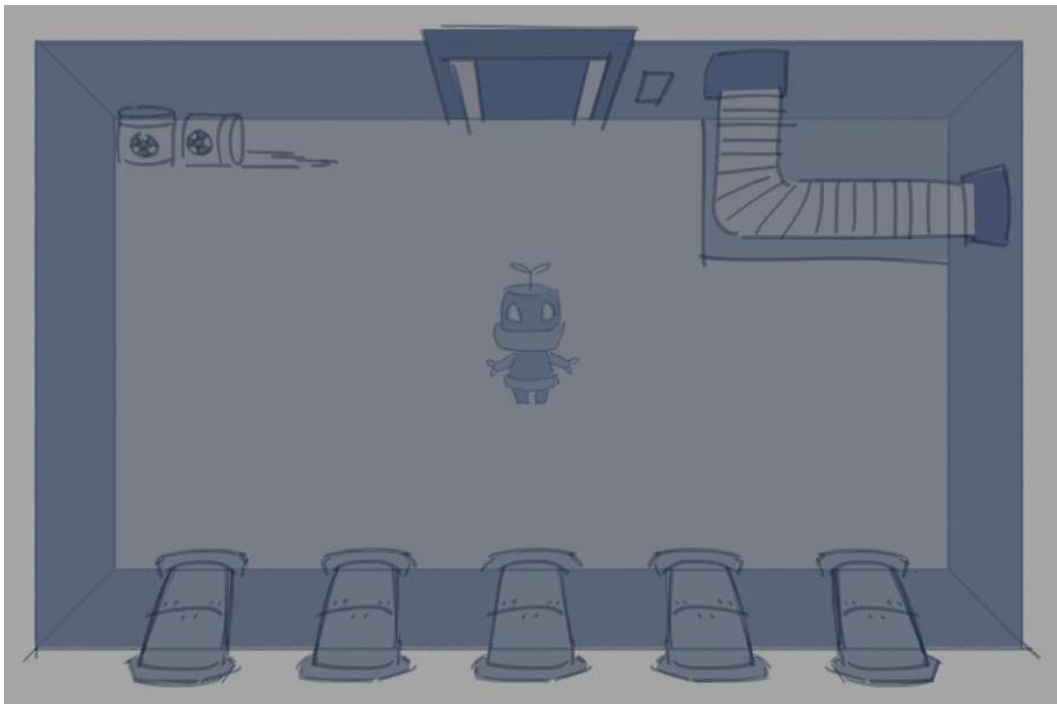
Upgrade UI. This will also be used as a base for NPC shop/trading UI as well.



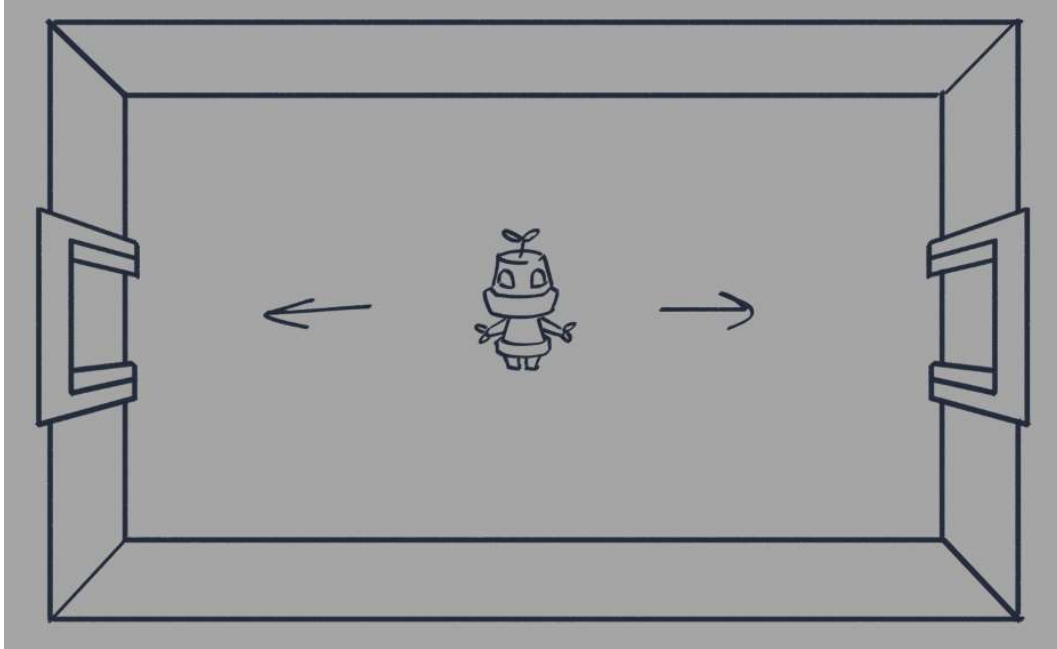
First stage. Lots of overgrowth.



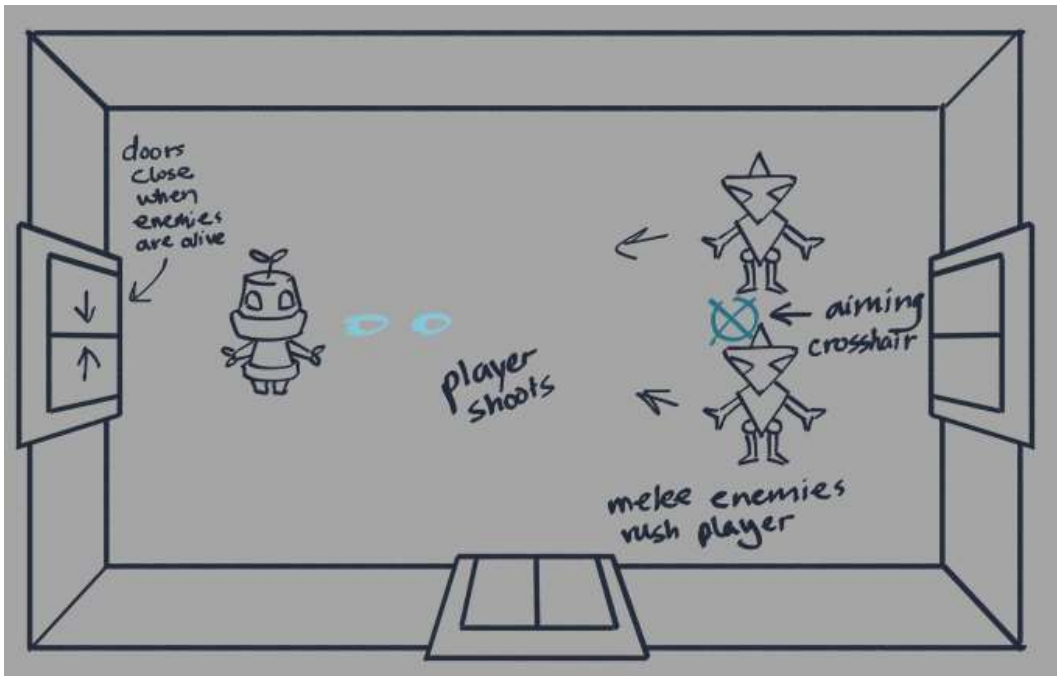
Second stage



Third stage. The stages get darker and more industrial as the player descends.



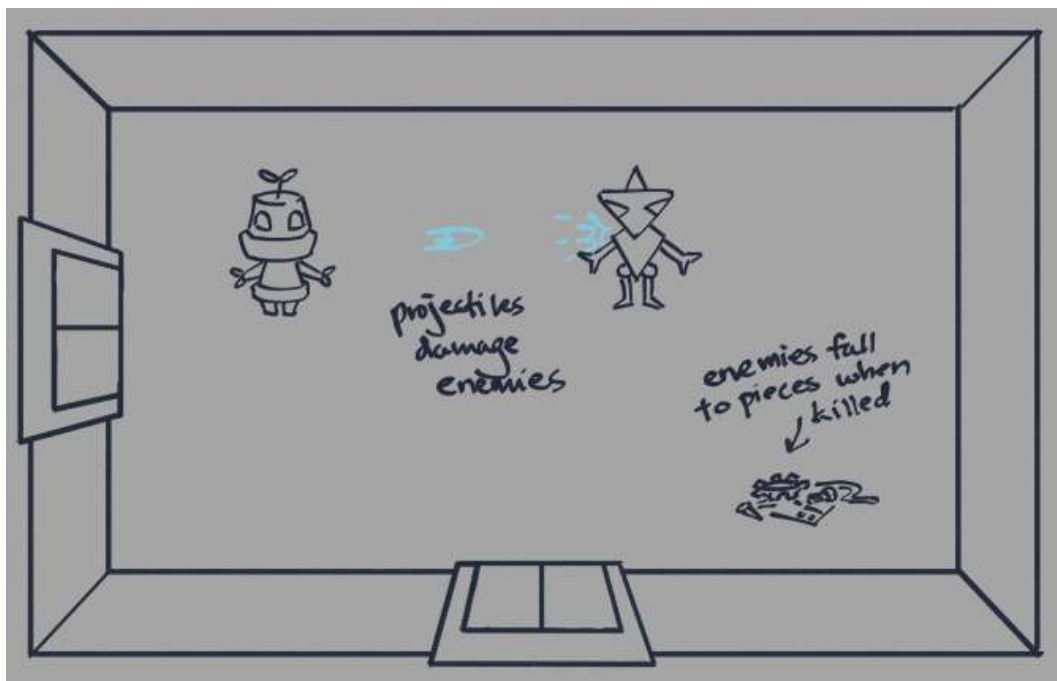
Players can choose which room to navigate to.



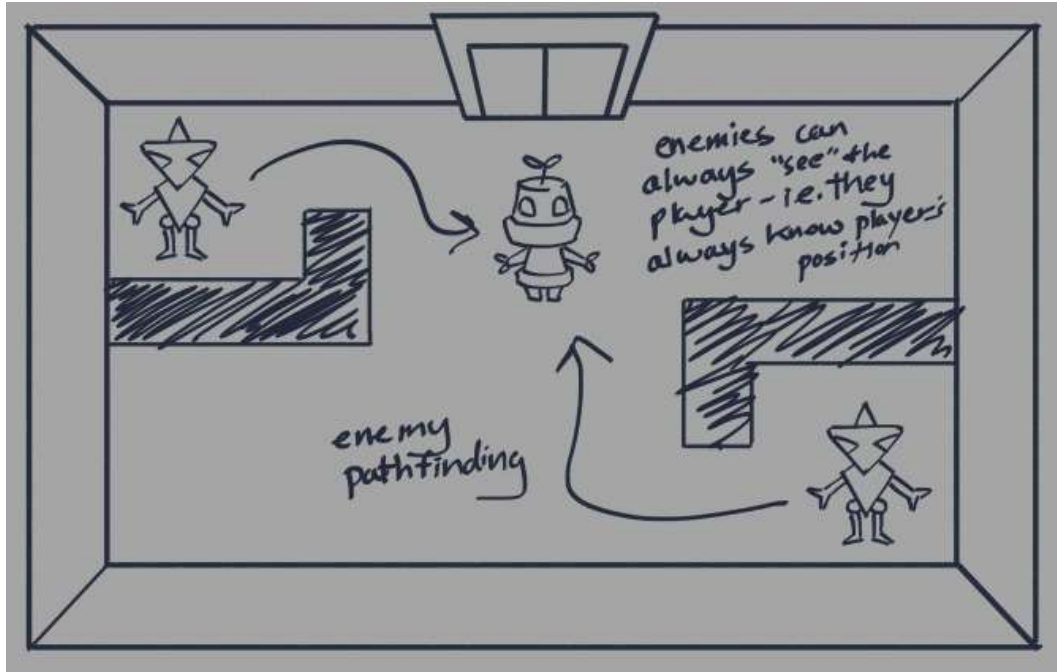
Doors close when enemies are present in a room. There is an aiming reticle that shows where the player's bullets go.



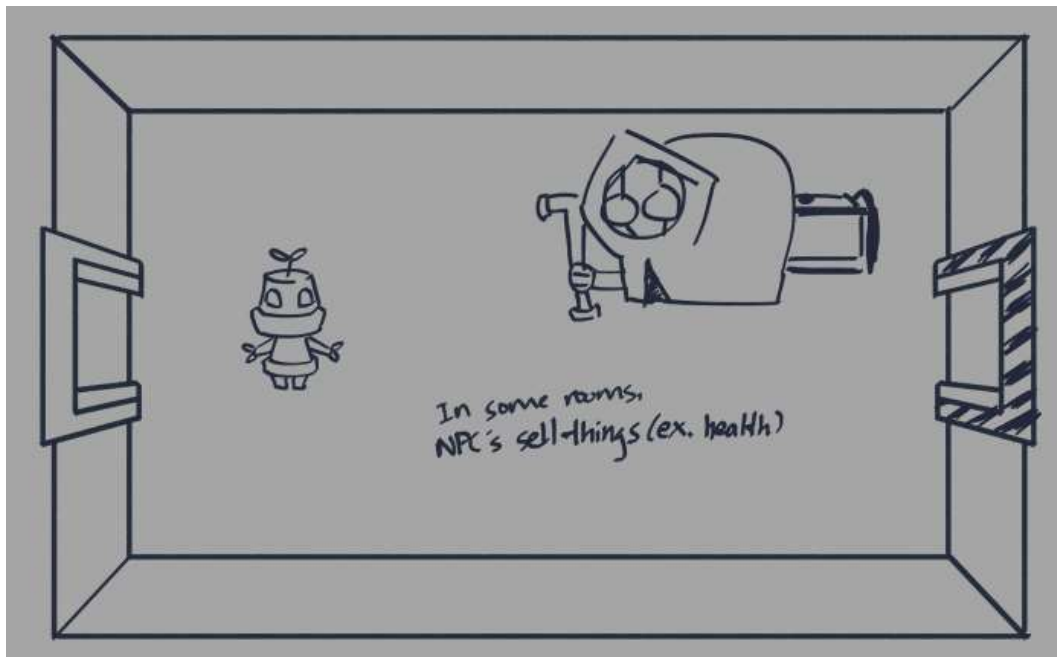
Basic enemy projectile and attack patterns.



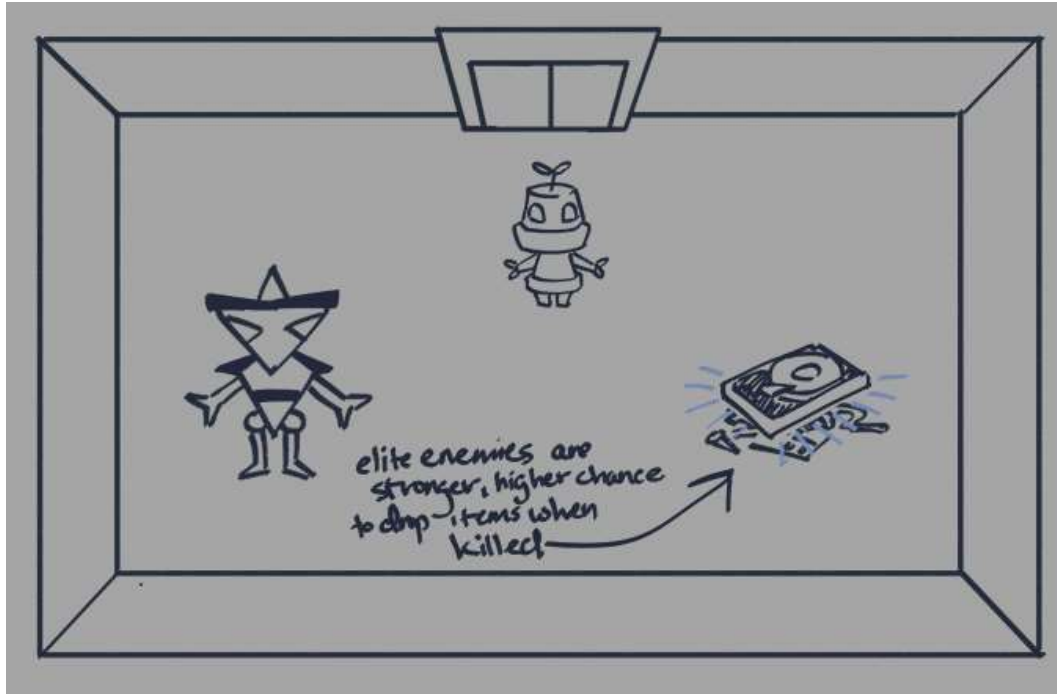
Projectiles damage enemies; dead enemies are reduced to pieces.



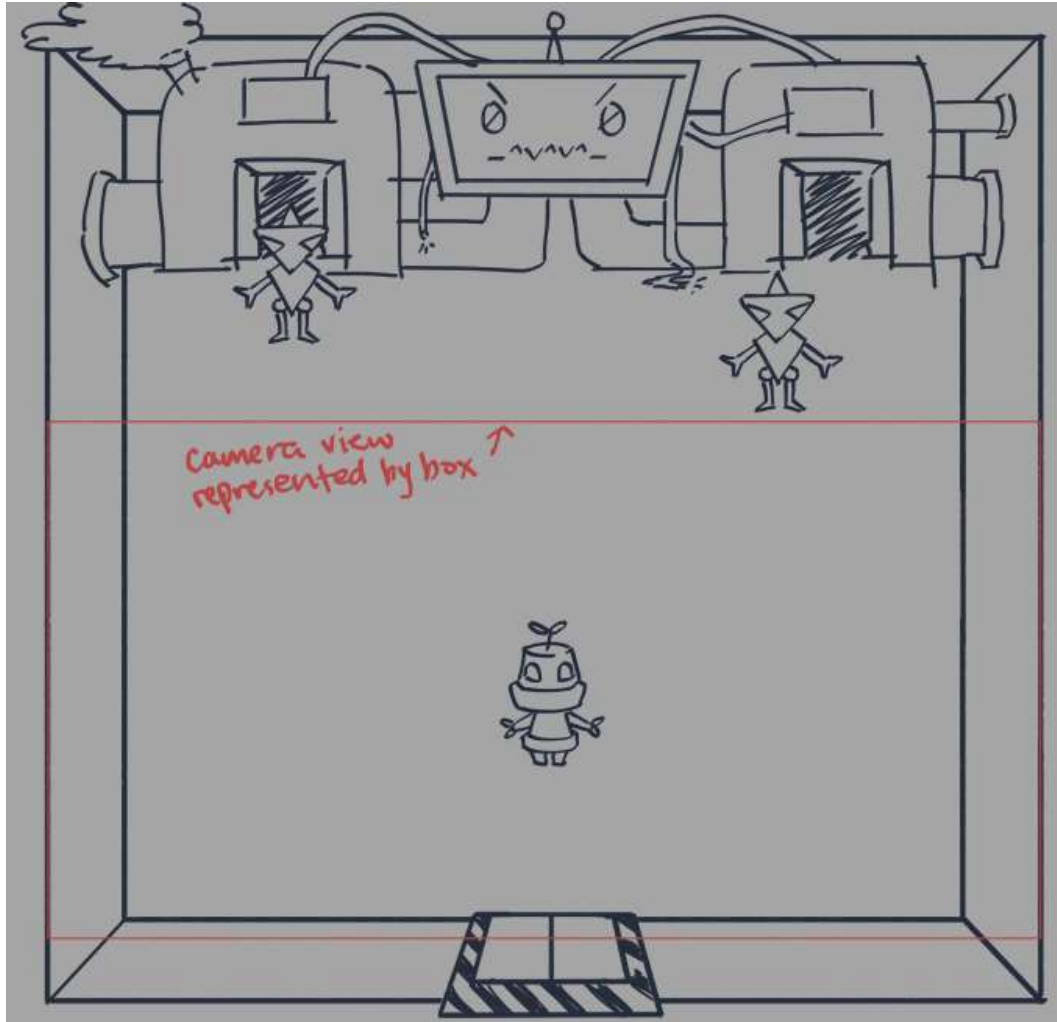
Basic enemy pathfinding. The enemies can always 'see' the player.



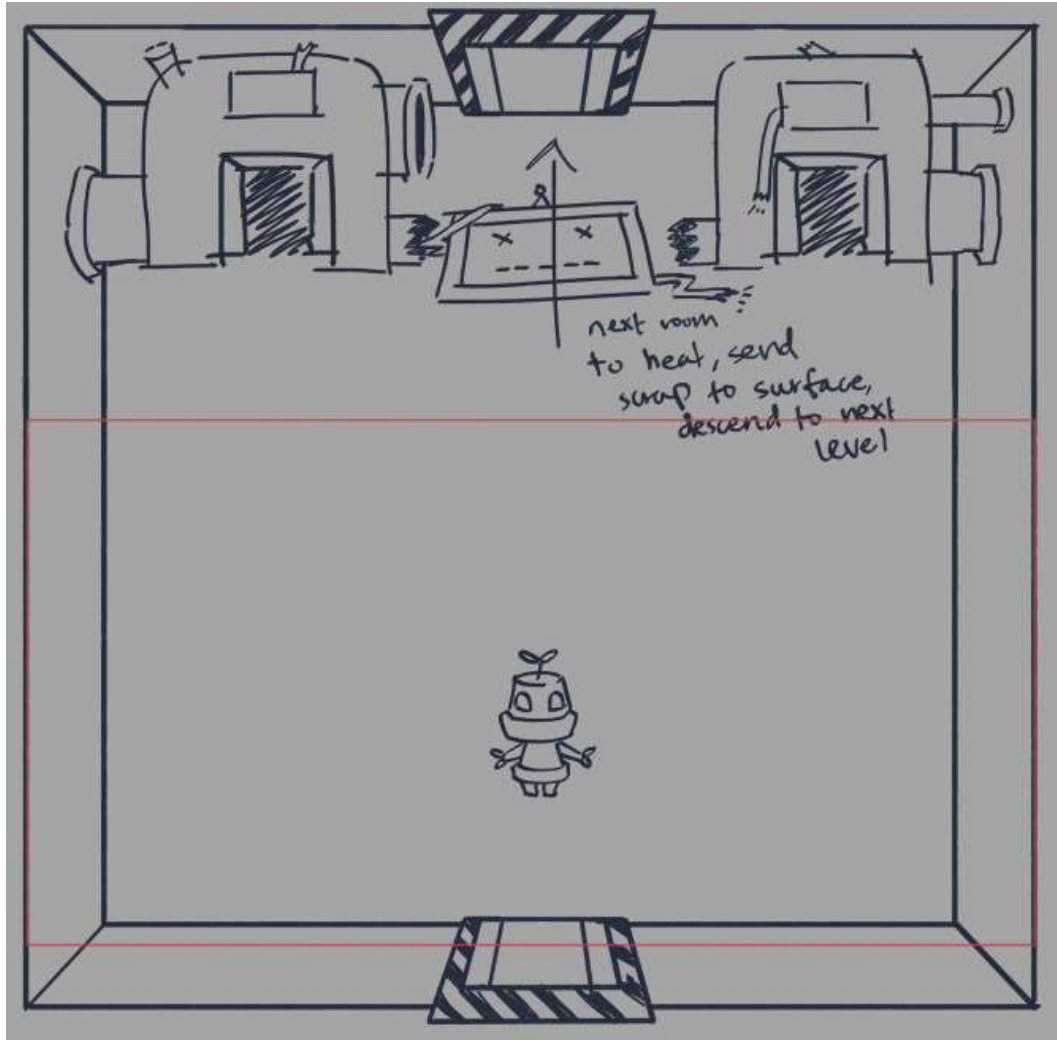
Some rooms will have NPC traders.



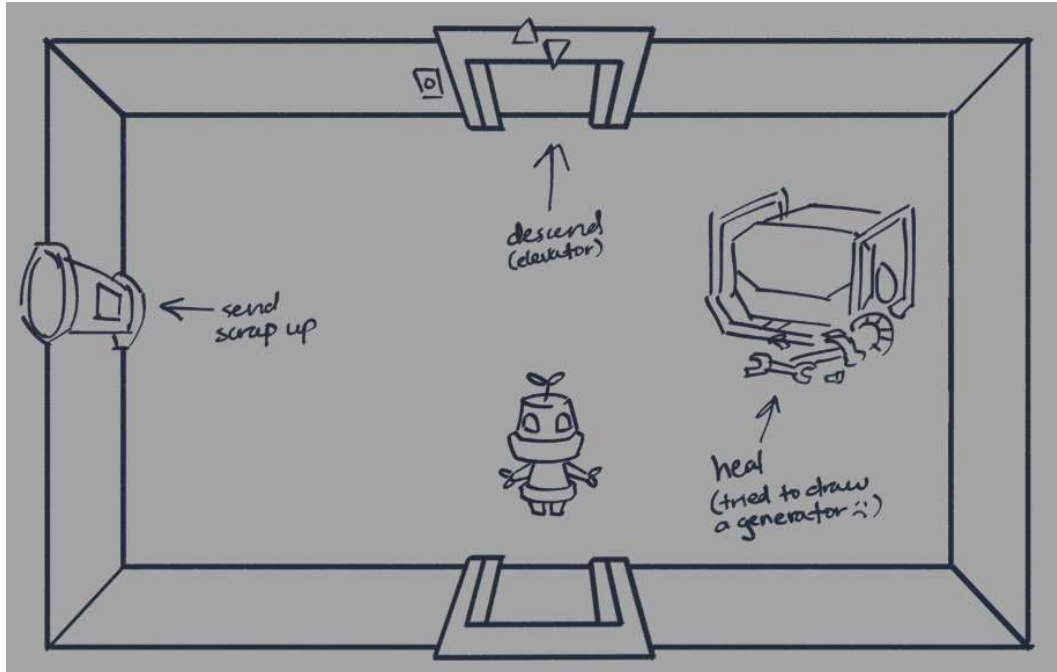
Elite enemies are stronger, w/ slightly altered sprites. They may drop more scrap, health, or even an item.



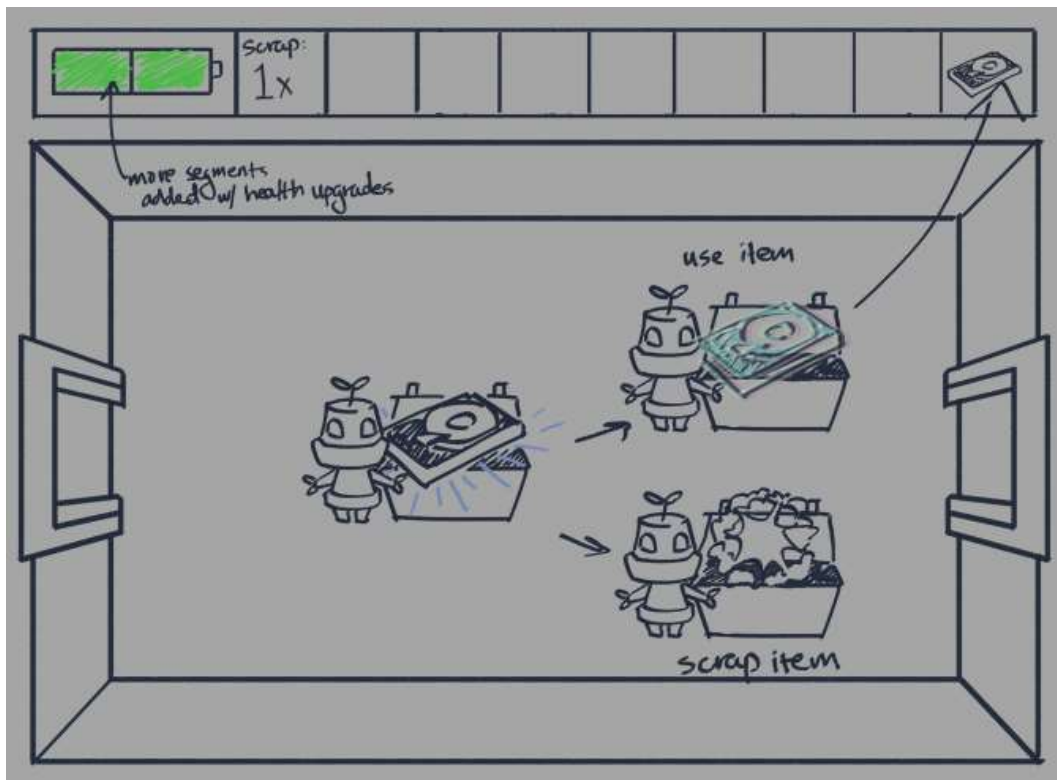
Boss fight concept. The room is larger than regular rooms, so the camera will track the player up and down.



After defeating the boss, a new room will open up.



The room that is unlocked by defeating the mother. The player has the ability to send scrap to the surface, heal up, and descend to the next stage.



Item pickup and item scrapping. Only 2 item slots will be unlocked at the start. The health bar will have more segments as the player finds or buys health upgrades.

Technical Elements:

Rendering: we will have 2D graphics representing the player (robot character) on a backdrop.

Assets: we will have background music, player and enemy sprites, and geometry for all visual assets.

2D geometry manipulation: we will have player and enemy sprites moving across the screen, and collision with projectiles, enemies, and walls.

Gameplay logic: the player will be able to interact with items (i.e. picking them up or scrapping them), and move around in the world. We will also have procedurally generated levels. We will also have NPC dialogue for traders and other NPCs. We will also have shop functionality (trader and permanent).

AI: we will have enemies that have basic and advanced pathfinding and have to decide when to attack, etc.

Physics: we assume that basic kinematics counts, which is satisfied by player and enemy movement, as well as by projectiles. We will also have collision functionality.

Advanced Technical Elements:

M1:

- We plan to implement a basic pathfinding for enemies in M1 where the enemies just move towards the player. (suggested feature 14, basic)
- We plan to implement particles from bullets, damage, and other sources will create particles that disperse randomly and experience air friction. Some enemies (such as bosses) and/or their weapons may emit particles which will also have a similar effect (suggested feature 8, basic)

M2:

- We plan to have basic integrated assets that we will create ourselves. (suggested feature 24, basic)
- We will add basic sound effects and background music. This includes upon shooting, swinging a weapon, taking damage, doors opening, dying, etc. (suggested feature 23, basic)

M3:

- We will add a simple rendering effect (suggested feature 1, basic)
- We will add complex geometry for the player, and mesh-mesh collisions for the player and projectiles. (M3 suggested feature, 'Complex Geometry')
- We plan to have reload capability, where the player's scrap (permanent currency), permanent upgrades to the mother, and game state within a run will be serialized and saved to persist after the game is closed. (suggested feature 19, basic)

M4:

- We will add story elements through dialogue boxes with NPCs and dialogue before or after the 3 boss fights (suggested feature 27, basic)
- We will also try to implement an advanced pathfinding system (Advanced decision-making) for enemies. This is a high priority feature that we would really like to include. It will provide the player with more of a challenge when going through rooms as enemies will chase the player to damage them in a smarter way (ex. try to dodge projectiles, move differently if at low health) (suggested feature 15, advanced)
- We will have enemies coordinate different attacks, such as moving to surround the player and changing paths based on how many other enemies are present. (suggested feature 17, advanced)

Devices:

Keyboard:

W - Move up

A - Move left

S - Move down

D - Move right

E - Interact (items, NPCs, etc.)

X - Scrap Item

Esc - Pause game/Open main menu

Mouse:

Movement - Aim weapon/Navigate menus

Left Click - Fire weapon/Select menu option

Tools:

- Libresprite for asset creation (time allowing)
- A text rendering library (maybe GLFont or FreeType)

Team management:

We will be doing our work in sprints, dividing the deliverables into user stories, deriving tasks from those stories, giving each task a time estimate, and prioritizing them before each sprint. Tasks will be assigned to a team member, aligning with the preferences we showed in our pitch video when possible. We will be using a task board to keep track of everything, including how long each task takes to finish (to improve estimates later on). We aim to finish deliverables 1-2 days before the actual deadline, so that we have time to double check and verify that all parts are working.

Development Plan:

A note on the development plan notation:

- R denotes that a task is required or implicitly required.
- S denotes that a task falls under suggested features.
- Where applicable, tasks are also annotated with the feature they correspond to on the Suggested Game Features page on Canvas, and whether they are advanced or basic (A or B, respectively).

Milestone 1: Skeletal Game

For milestone 1 [M1], you should have a basic version of your target game. It should incorporate basic rendering, input-driven response, 2D motion, correct basic collision handling, event-driven/random response, and a minimal set of assets. Feel free to use the assignment 0 and 1 templates as a starting point for your game code.

Week 1

UI

Placeholder for UI/HUD elements - **R**

- render space for UI (item slots, health, scrap counter)
- 1-2 hrs
- display player health once it's added - KELCY

Utility

Add interaction functionality - **S27B** (buildup to M3)

- make 'interactable' component (will be used by npcs and items)

- make a system for interactables that calls a fn when an interactable is interacted with
 - maybe also have a check to make sure the interactable is in range of the player
 - 1hr (ECS will already be in place) - KELCY
- ^ this will initially be used for items in M2, and later, for dialogue in M3*

Gameplay Mechanics

Implement basic graphics for player, enemy, and floor- **R**

- make entities for each
- 2hrs - KELCY

Add functionality for player movement from keyboard input for each direction - **R**

- relies on graphics task in UI
- <1hr - ARSHVIR

Add functionality to take user input for LMB to fire and mouse movement for aim - **R**

- make bullet
- 3hrs - ARSHVIR

Add functionality to track player and enemy health - **R**

- common component 'health'
- no death yet
- display player health
- 1hr - ARSHVIR

^ health will be a small number of hit points for enemies and players (all entities starts with 2)

AI

Implement pathfinding system for enemy to track player - **S14B**

- 1-2hrs - PATRICK

Geometry/physics

Create a functional hitbox for the main character to collide with solid objects (walls, enemies, etc.) - **R**

- create hitbox component
- add system for collision detection
- add system for blocking entity movement using collision detection system
- add component indicating whether an entity blocks movement
- 2hrs - ASHER

Create reusable wall objects with collision - **R**

- 1hr - ASHER

Implement collision and hitbox for enemy - **R**

- 1hr (reuse system that player hitbox uses) - ASHER

Add a hitbox to bullet and functionality to destroy bullet on collision with solid object/enemy - **R**

- 2hrs - ASHER

Art/Sprites

Apply interpolation to character's movement - **R**

- 1hr - MATT

Apply interpolation to enemy's movement - **R**

- 1hr - MATT

Apply interpolation to bullet's movement - **R**

- 1hr - MATT

Testing

Develop test plan - **R**

- outline testing strategy
- 2-3hrs - PATRICK

Week 2

Utility

Gameplay Mechanics

Implement damage application on player by enemy - **R**

- 1-2hrs - PATRICK

Implement damage application on enemy from bullet - **R**

- 1-2hrs - PATRICK

Add basic death handling for player (just full reset) - **R**

- relies on health component being implemented
- 1hr - PATRICK

Add simple death handling for when enemy HP is reduced to 0 (just disappear) - **R**

- 1hr - PATRICK

Geometry/physics

Create reusable floor objects - **R**

- background floor object (image) AND random floor decorations
- floor decor may block player movement -> collision needed
- use 'blocking' component
- 1hr - MATT

Well-defined game-space boundaries (at least one room) - **R**

- build out a room (using wall and floor objects from earlier)
- 1hr - MATT

Add particles appearing from the main character when damage occurs which have air friction - **S8B**

- ex. little sparks
- 2-3hrs - MATT

Art/Sprites

Add texture to all geometry for walls, character, objects, and enemy - **R**

- choose pixel size?
- 2-3hrs - KELCY

Testing

Testing/optimizing, keep track of bugs and fixes - **R**

- 3hrs - ARSHVIR

Other

Create demo Video - **R**

- 2hrs - everyone

Milestone 2: Minimal Playability

For this milestone, you should continue to support all required skeletal game features and fix any game-stopping bugs (P0 and P1). You should augment those features with core gameplay logic, incorporate additional assets and play elements that allow for non-repetitive gameplay, fix all known bugs, and perform playability testing. There should be 2 minutes of non-repetitive gameplay (i.e. multiple rooms)

Week 1

UI

Implement main menu functionality (on start) - KELCY (4-5hrs)

^ note that this doesn't include implementing the button actions, just make them clickable and make them output something to use later

Buttons labeled with **R** MUST HAVE FULL FUNCTIONALITY

- Start button - **R**
- Test room button
- Help button - **R**
- Save & Quit button
- Credits Button - **O**
- Shop button
- may need to have createButton fn
- may need to move createWindow to render system
- **create** UI/menu system
 - functions for textboxes, buttons, etc.
 - consider changing mouse cursor when hovering over button

Implement help screen - **R** - KELCY (<1hr)

- static image showing controls + mechanics

^ Plan B: have a basic graphic in the first room showing keybindings and mechanics, i.e. how to move, shoot, use items, scrap

~~Add inventory tab on HUD~~

Utility

Add a test room (big room) with functionality within code to add enemies - **O**

Develop algorithm for procedural generation of a floor - **R**

- 3-4hrs
- ASHER

Geometry/Physics

Create reusable floor objects - **R**

- Random floor decorations
- floor decor may block player movement -> collision needed
- use 'blocking' component
- 1hr
- MATT

Make enemies and players push each other away (collision handling)

- ASHER (10 mins)

^ note for whoever does this, enemy-enemy collisions are not currently being added to the collisions registry, you'd need to add it to the physics system's checkCollisions

Gameplay Mechanics

Add functionality for item drops from enemies (probability-based)

- ARSHVIR (2 hr)

Implement ability to pick up items (max 2, not changeable for now)

- ARSHVIR (2 hr)

Implement one item of each type

- speed, damage, rate of fire, bullet lifetime/range, health pack
- Modifier component maybe
- ARSHVIR (2 hr)

AI

Implement basic decision tree for two AI enemies (melee and ranged) - **R**

- enemies make decisions based on player's actions/movement
- no and type of enemy spawning
- PATRICK

Art/Sprites

Implement basic assets for items - **R**

- MATT (PATRICK assist) (1-2hrs)

Implement basic assets for new enemies - **R**

- MATT (1hrs)

Create dynamic title asset

- already "done"; might need tweaking
- will be used in main menu
- KELCY (10-20mins if changes needed)

Bugs/Testing

Fix bugs

- ASHER

Develop test plan

- ARSHVIR (2 hr)

Week 2

UI

*Create overlay for FPS counter - **R**

- *basic functionality done; maybe consider doing overlay
- ARSHVIR (1 hr)

Implement pause menu - all buttons require complete functionality

- Resume button
- Save & Quit button
- Help button?
- ASHER (1+hrs)

Update health in UI dynamically - KELCY (1hr)

Update items in UI dynamically - KELCY (0.5hr)

Update scrap in UI dynamically - **O** - KELCY (1hr)

Add permanent upgrade shop screen - full functionality will be done in GAME

MECHANICS tasks

- make sure to link UI to actual game elements
- use the UI system from main menu (like world system, with init and all)
- MATT (2hrs)

Utility

Add reload capability (saving permanent upgrades and game state) - **S19B**

- save and load from JSON file
- PATRICK

Add save & quit functionality - **S19B**

- link save and load functionality to buttons in UI
- PATRICK

Add pausing functionality - **R**

- 2hrs
- ASHER

Geometry/Physics

Implement mesh-based collision - **R**

- ARSHVIR + ASHER
- 1hr each

Bug fixes where necessary

- ASHER

Gameplay Mechanics

Implement item effects on player - **R**

- ARSHVIR (1 hr)

Implement dying (shows game-over screen (- **O**), returns to title screen) - **R**

Add option to scrap item and increases player's scrap count

- scrap count shown in in-game UI is scrap collected during run
- PATRICK

Implement doors and room switching (in renderer and world system)

- ASHER (2hrs+)

AI

Bug fixes where necessary

Art/Sprites

Add damage sound effect - **S23B**

- ARSHVIR (1 hr)

Add shooting sound effect - **S23B**

- ARSHVIR (1 hr)

Add melee sound effect (enemies) - **S23B**

- ARSHVIR (1 hr)

Add basic background music - **S23B**

- *2 tracks, one for combat and one for post-combat*
- ARSHVIR (1 hr)

Implement basic attack animations for the two enemies and player - **R** - KELCY (2.5-3hr)

- player firing: particle effects or explosion
- enemy melee: swing animation
- enemy firing: particle effects or explosion

Bugs/Testing

Execute tests and fix bugs

- MATT (2 hr)

Update test plan

- Patrick

Other

- M2 Video - MATT (2hr)

Milestone 3: Playability

For this milestone, you should have a completely playable game, continue to support all required features from prior milestones, and have fixed the majority of your latent bugs. In addition, you should now also include advanced features such as detailed geometry, non-linear motion, and time-stepping based physics. Your game should provide robust continuous play with no memory leaks, crashes or glitches.

Key to this milestone is that you test the playability of all new features and ensure alignment with your team's game development plan. You are encouraged to ask people outside your team to playtest your game, so you can fine tune your gameplay and interface.

Week 1

Required:

- 5 mins gameplay (should be all new content)
- No mem leaks
- Robust user input (tabbing out, random keys, etc)
- No input lag, stuttering, framerate drops
- No bugs
- Creative component: 40%
 - Reloadability - 10%
 - Story components - dialogue 10%: Asher
 - Geometric assets 20% - Matt

UI

Simple rendering special effect (when the player gets hit, when enemies die, while collecting items) - **S1B** - KELCY

Implement text boxes - **S27B** - ASH

Add tooltips for item descriptions (hover over item with cursor)

Continue button on main menu - KELCY

- functionality done by pat

Utility

Change rendering system to use z-buffering - KELCY

Add NPC room - **R** ASH

Add a previous player death room to game - **S27B** - ASH

Add boss room - ARSH

Add 1 more stage

^ currently, they will all be the same apart from their final boss

Serialize player deaths for previous player death room - **S27B** - PAT

Gameplay Mechanics

Implement peaceful room (NPC, previous death) framework - **R** - ASHER

Implement 5-10 items - O - PAT

Implement 2 more unique enemy types with their own attack patterns - PAT

Add working dialogue from NPCs. - ASH

^ New enemies will complement old enemies on newer stages, and they will have more health and more complex attack patterns

Make items scrappable from ground - PAT

Art/Sprites

Design previous player death room - KELCY

Design/source assets for 5-10 items - O - KELCY

Design/source assets for 1-2 enemies - MATT

NPCs:

- Design/source assets for 1 NPCs - **S27B** - Arshvir
- Design the story for 1 NPCs - **S27B** - Asher
- Design the rooms for 1 NPCs - **S27B** - Matt

Bosses: - KELCY

- Design/source assets for 2 bosses - **R**
- Design the story for 2 bosses - **R**
- Design the rooms for 2 bosses

Bugs/Testing

Execute tests and fix bugs

Week 2

Utility

Implement advancing between stages - **R** - PAT

Save dialogue state for npcs - PAT

Gameplay Mechanics

Implement boss attacking pattern - **R** - ARSHVIR

Implement function to talk to a npc - **R** - ASH

Implement dropping items on the ground - **R** - PAT

Design the attacking style, attack pattern for final boss - **R** - ARSH

Add functionality for a basic shop from the menu screen with three upgrades - KELCY

- add item slot
- upgrade base damage
- critical strike chance - **O**
- dodge chance - **O**
- upgrade max health
- ^add UI as well

Art/Sprites

Design/source the asset for final boss - **R** - KELCY

Design the story for final boss - **R** - KELCY

Design the room for final boss - ARSHVIR

Design/source assets for 2 NPCs - **S27B** - KELCY

Design the story for 2 NPCs - **S27B** - ASHER

Design the rooms for 2 NPCs - **S27B** - ASHER

Write dialogue exchange tree for 3 NPCs (dialogue from NPCs, and dialogue option from player) - **S27B** - ASHER

Write dialogue exchange tree for 3 bosses (dialogue from bosses, and dialogue option from player) - **S27B** - ASHER

select and add background music for regular room/boss/npc - **S26A** (buildup for M4)

write script for winning cutscene - **R** - ASHER

Add sfx for dialogue (like undertale dialogue sound effects) - ASHER

Implement boss fight music - **O** ARSH

Bugs/Testing

Execute tests and fix bugs

test plan - PAT

Milestone 4: Final Game

The final version of your game should support robust and continuous gameplay and integrate advanced game elements. It should be self-contained, i.e., players should be able to play the game with no outside help or instructions other than those provided by the game itself. You should implement one or more additional advanced gameplay features (see below) and you are welcome to incorporate advanced features using third-party libraries (verify with TA). Your final video will demonstrate your final game, highlight the advanced features, and provide a review of your game's development and evolution.

Key to this final milestone is that you test the playability of all new features and ensure alignment with your team's game development plan. You are encouraged to ask people outside your team to playtest your game, so you can balance and fine tune your gameplay and interface.

- A* HERE

Week 1

UI

Ensure resolution and aspect ratio are consistent across different machines/displays - **R**

- EVERYONE

Ensure all inputs are appropriately handled, including player control, item use, pause, switching windows, minimizing the window, hitting invalid inputs. - **R**

- EVERYONE

Add tutorial page button in pause menu - **R** - ASHER DONE

Add proper health bar UI - KELCY

Add crosshair indicator for shooting cooldown - **ARSHVIR**

Add map - **Matt**

Player death screen (enemies killed, scrap gained, etc.) - **ARSHVIR**

- textbox over death screen

- at least: # enemies killed and # scrap gained

Increase window size - KELCY

Utility

Work on ensuring no crashes, glitches, lack of smoothness. The game should run consistently and as expected. - **R**

Fix any memory hoarding - **R**

Fix any memory leaks - **R**

If necessary, create additional layout options for procedural generation to ensure gameplay variety. - **R**

If necessary, create additional enemy types to ensure that enemy generation is sufficiently varied to produce fresh gameplay. - **R Matt**

Gameplay Mechanics

Finalize interactive components based on the implementation of most other mechanics (eg. player speed is not too slow to get away, not so fast that enemy pursuit is trivial). - **R**
Get rid of melee damage from player (i.e. player should not inflict melee damage) - **Matt**
Get rid of melee damage from any ranged enemies - **Matt**
Dialogue before boss fights - ASHER
Add indicator (maybe sparkles?) to items to show that they can be picked up - KELCY
Item tooltips - ASHER
loot room - **O**
5-10 new items - **ARSHVIR**
Third Boss fight - **ARSHVIR**
doors - ASHER

- should "close" when in combat
- update hitboxes

AI

A* pathfinding. - **S14A** - PATRICK

Art/Sprites

If necessary, design/source assets for additional enemy types - **R - Matt**
Add dialogue tree options and add more dialogue - **S27B** - ASHER
Boss death sprites/animations - **Matt**
Player death animation - **Matt**
Remove active item instructions from help screen - KELCY
Make different wall and floor assets for stages 2 and 3 - KELCY
Make assets for new items - KELCY
NPC idle animations - KELCY - **O**
Add different music for stages 2 and 3 - ASHER

Bugs/Testing

Fix issue with scrap added from a run not being usable for purchasing - KELCY
Investigate issue where player dodge indicator (flashing blue) does not appear - KELCY
Extensive playtesting by ourselves and others to hunt for unexpected bugs and ensure all components are fully operational and that the game can be played without prior knowledge. - **R** - EVERYONE

Week 2

UI

After all implementations, ensure that the game runs in real-time (no lag or jumpiness, consistent frame rate). - **R**
Conduct performance testing to prepare for our discussion of it in the demo video. - **R**
Finalize the tutorial page from the main screen to cover all aspects of the game. - **R**

- e.e.g tooltips on items + explain hover on items
- explain doors lead to other rooms
- KELCY

Hover crosshair over buttons should change crosshair - KELCY

Utility

Fix any remaining crashes as discovered from ongoing playtesting. - **R**

Geometry/Physics

Finalize new and existing physics systems, ensure no bugs remain. - **R**

Gameplay Mechanics

Ensure polished tutorial concisely presents game instruction to a player with no prior knowledge based on previous week's playtesting. - **R**

Procedurally generated rooms - ASHER

Balancing:

- Balance player and enemy health
- Balance items
- Balance number of rooms
- Increase damage for heavy enemies
- increase number of enemies per room
- decrease number of rooms in a stage
- decrease enemy size

AI

Bug fixes. - **R**

Art/Sprites

Last chance to fix any other lasting bugs. - **R**

Proofread dialogue - **S27B**

Bugs/Testing

Fix any remaining bugs in the project, by order of priority if they can't all be done. - **R**

Track these bugs for reporting. - **R**